LinkedIn

Ricardo A. Sanchez

Madison, WI 53705

rsanchez24@wisc.edu
787-590-9594

Education

- **University of Wisconsin** — Madison, WI
  PhD in Biomedical Engineering — *June 2027*
  MA in Biomedical Engineering; GPA: 3.75/4.00 — *August 2024*
- **University of Puerto Rico** — Mayaguez, PR
  BS in Mechanical Engineering (Cum Laude); GPA: 3.40/4.00 — *December 2023*

Experience

- **University of Wisconsin – Ludwig Lab** — Madison, WI
  Graduate Researcher (Advisor: Dr. Kip Ludwig) — *Jun 2022 – Present*
  - **Project: PS-OCT Guided Injectrode System (R01 Grant)**: Leading a translational project to improve posterior tibial nerve stimulation (PTNS) for Overactive Bladder (OAB) by enhancing targeting precision and minimizing side effects.
    - Designed and validated a PS-OCT-guided needle probe for minimally invasive electrode/drug delivery to sub-tissue targets.
    - Conducted $n = 10$ porcine studies; analyzed EMG and ECAP data to evaluate on-/off-target stimulation across implantation depths.
    - Developed 3D computational modeling pipeline using microCT, MUSE imaging, and histological segmentation to guide electrode placement.
    - Collaborated across imaging, surgical, and analysis teams to integrate modeling and preclinical insights into device development.
  - **Project: Electrochemical Safety Standards for Neuromodulation (Safety Grant)**: Establishing safe stimulation limits and design parameters for implantable electrodes through benchtop and in vivo characterization.
    - Performed electrochemical impedance, voltage transient, and cyclic voltammetry testing on electrodes of varying size/materials.
    - Built and validated a longitudinal test platform for chronic safety evaluation of pulse generators in large animal models.
    - Developed a frame work to understand the relationship between electrode geometry and the requirements to efficiently use the available geometry for safe charge injection.

Boston Scientific — Dorado, PR

- *Mechanical Engineering Capstone – 4Front Stake Cable to Fitting Project* — *Jan 2022 – Jun 2022*
  (Advisors: Dr. David Dooner & Jorge Soto)
  - **Project: Manufacturing Process Redesign for RELIANCE 4-FRONT™ Lead**: Led a process improvement initiative to reduce defect rates and improve throughput in a high-volume medical device manufacturing line.
    - Designed and deployed a motorized blade fixture to replace manual cutting operations, reducing operator-dependent variability and material waste.
    - Implemented in-line fault detection to identify process deviations early and prevent defective production batches.
    - Delivered a 50% reduction in dead time and reduced annual manufacturing costs by $\approx$ \$422,328.

Skills

- **Quantitative Skills**: Large Animal Models, Electrochemical Analysis, TDT, OCT Imaging, Linear Optimization, Risk Analysis, Structural Analysis, Fabrication Technologies
- **Computer Skills**: COMSOL, Synapse, Ascent, Neuron, Python, Julia, CAD, Eagle
- **Language Skills**: Spanish and English